

AKSHAT MIDHA

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EDUCATION

SRM Institute of Science and Technology

2024 – 2027

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

TECHNICAL SKILLS

Languages

- Python, C++, JavaScript, SQL

Frontend

- React.js, Next.js, Tailwind CSS

Backend

- FastAPI, Flask, Node.js, Express.js

Databases

- PostgreSQL, MongoDB, Redis, SQLite

DevOps

- Docker, Kubernetes, Nginx
- GitHub Actions, Prometheus, Grafana

Cloud

- Vercel, Railway, Render

Tools

- Git, SQLAlchemy, Alembic
- Socket.IO, RabbitMQ, JWT, REST APIs, System Design

PROJECTS

WireDesk - Custom Wire Config & Production Display

[Github](#)

- Designed a real-time production order management platform using **Flask** and **PyQt** with synchronized desktop clients over WebSockets.
- Engineered backend integrating **PostgreSQL**, **Redis Pub/Sub**, **SQLModel**, and **Alembic** for reliable state management and schema versioning.
- Dockerized the application stack and configured **Nginx** with a **PostgreSQL** primary-replica database architecture to enable scalable deployment and high availability.
- Implemented **Redis-backed caching** and **real-time event broadcasting** to efficiently synchronize manufacturing updates across connected terminals.
- Built a production-ready architecture incorporating monitoring, logging, and modular services for maintainability and scalability using **PgBouncer**, **Grafana**, and **Prometheus**.

TerraSense AI - Crop Recommendation Platform

[Github](#)

- Developed a backend platform that combines satellite imagery and environmental data to generate crop recommendations using **Google Earth Engine**.
- Built asynchronous **FastAPI** services to process NDVI extraction and NASA POWER weather data for scalable inference workflows.
- Designed normalized database schema with **SQLAlchemy** and managed schema evolution using **Alembic** migrations for **PostgreSQL** database.
- Containerized services with **Docker** and authored **Kubernetes** deployment manifests supporting horizontal scaling of worker nodes.
- Integrated an end-to-end inference pipeline with REST APIs, background processing, and persistent storage for prediction history through **Redis**, **RabbitMQ**, **Celery** and **Nginx**.

TalentIQ - Career Platform

[Domain](#)

A scalable, microservices-driven career platform featuring a server-side rendered front-end, secure authentication, AI-powered resume analysis, job recommendation engine, and robust payment integration

- Architected a microservices-based career platform separating authentication, job processing, user management, and payment services using **Express** and **Node.js**.
- Developed a responsive server-side rendered front end using **Next.js** to improve SEO and application performance.
- Built secure REST APIs and service-to-service communication to support scalable distributed application workflows using **Redis** and **Kafka** along with **Neon PostgreSQL**.

- Implemented secure authentication, authorization, and payment workflows while maintaining consistency across distributed services.